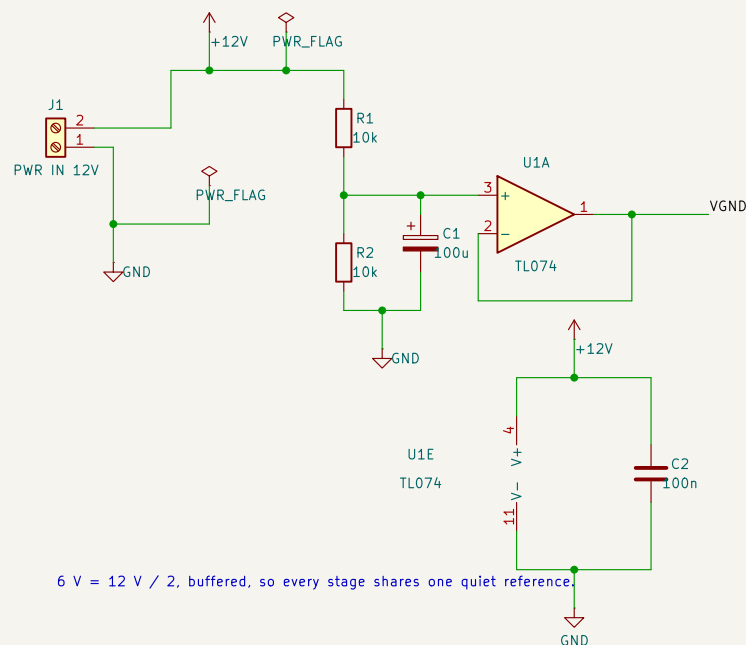
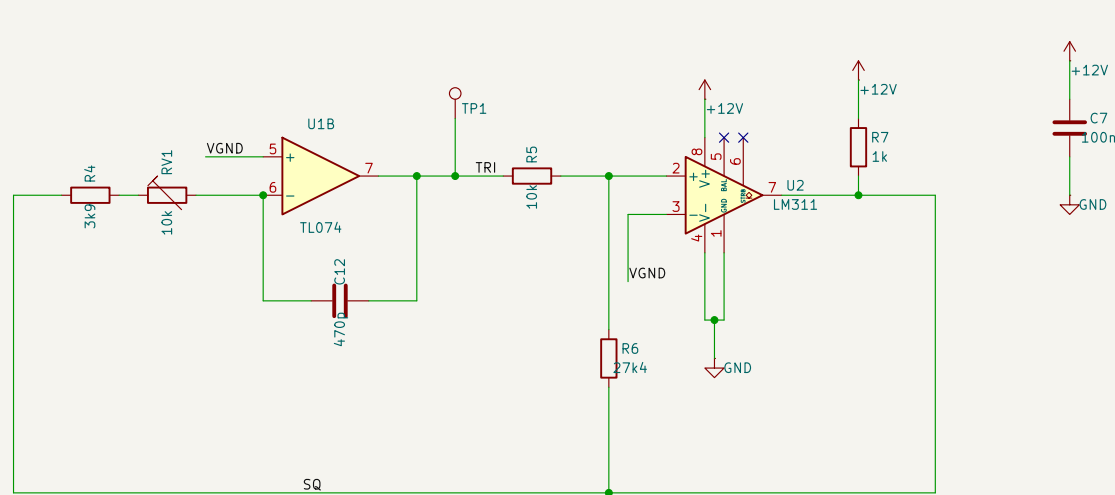


A Power in and 6 V virtual ground



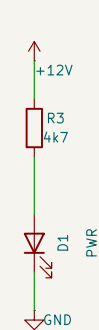
6 V = 12 V / 2, buffered, so every stage shares one quiet reference

B Triangle carrier ~250 kHz (LM311 Schmitt + TL074 integrator)



$f = V_{sw} / (8 R C)$: $R_4 + R_{V1} \sim 5k\Omega$ with $C_{12} 470p$ gives 250 kHz; R_5/R_6 set the ± 2 V window about 6 V

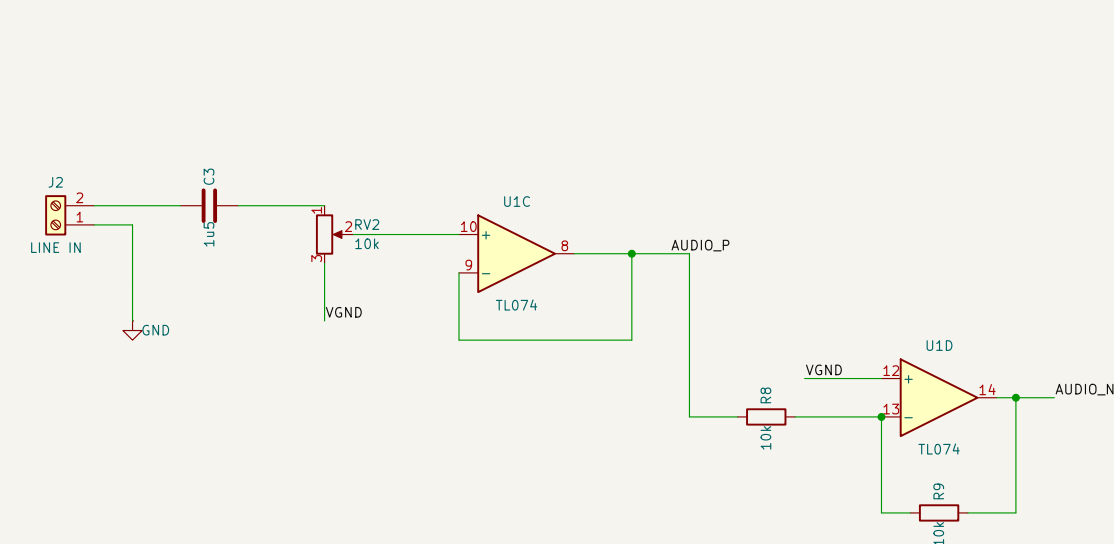
H Indicator and bring-up



Test points TP1..TP5 are solder lugs on TRI, PWM_A, both switching nodes and the filtered output.

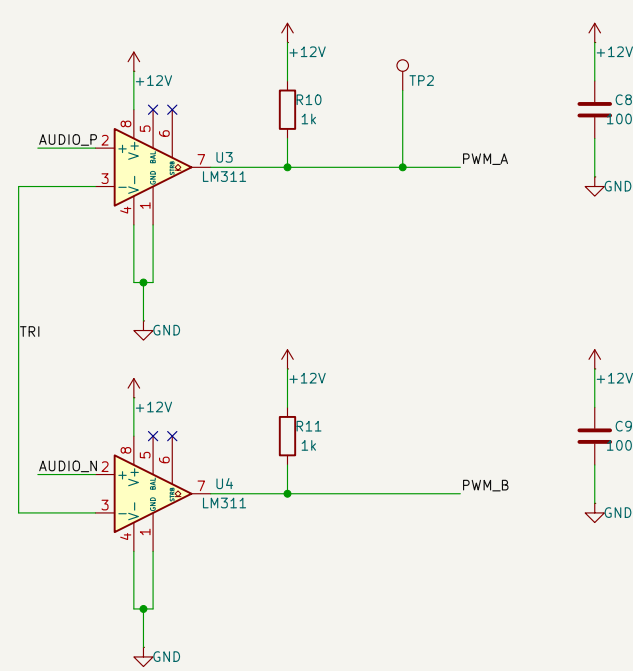
Bring up with NO MOSFETs fitted: check 12 V and 6 V, then the triangle, then both PWM outputs, then the HIP4082 dead time on a scope. Only then fit Q1..Q4, on a current-limited supply into a dummy load.

C Input stage (level $\rightarrow$ buffer $\rightarrow$ inverter)

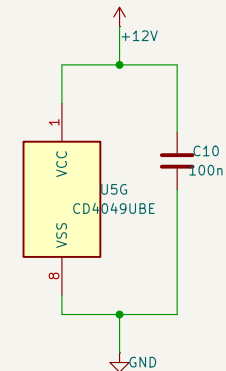
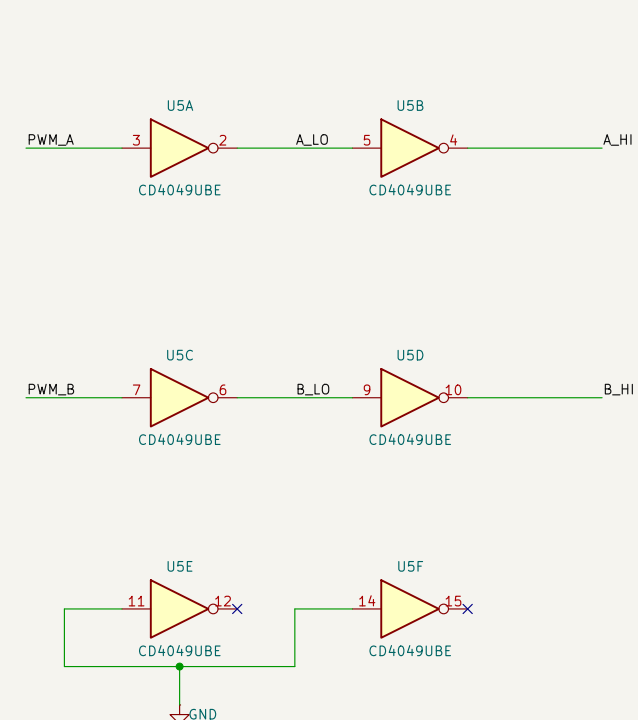


AUDIO_P and AUDIO_N are equal and opposite about 6 V – that is what swings the bridge both ways.

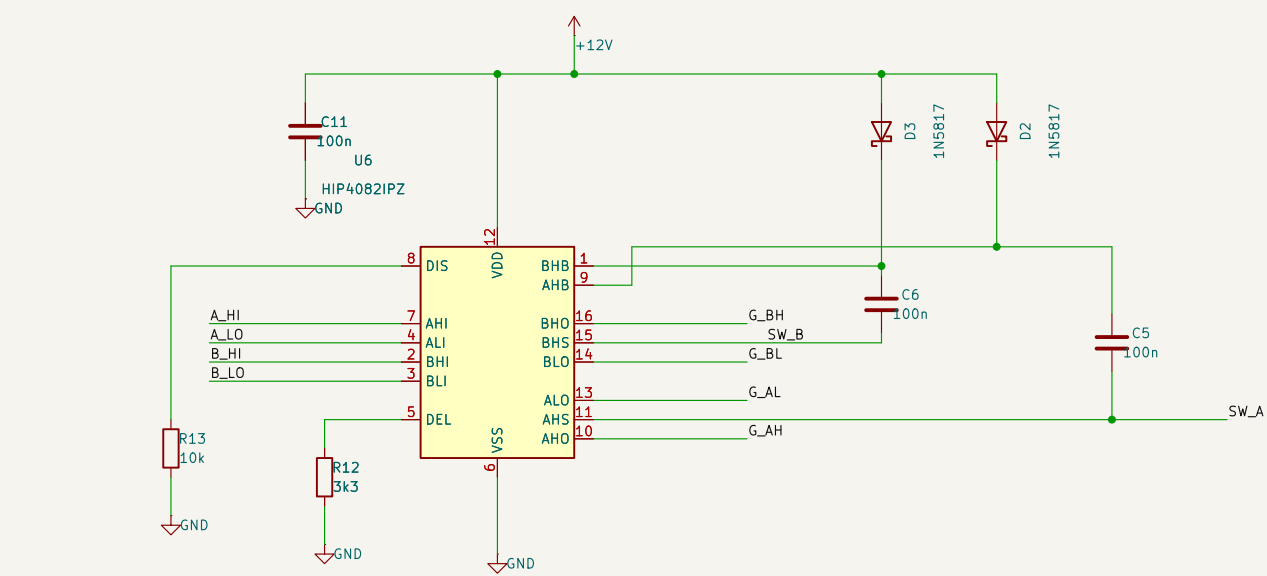
D PWM comparators (audio vs triangle)



E1 Complement generation (CD4049)



E2 Gate driver HIP4082IPZ



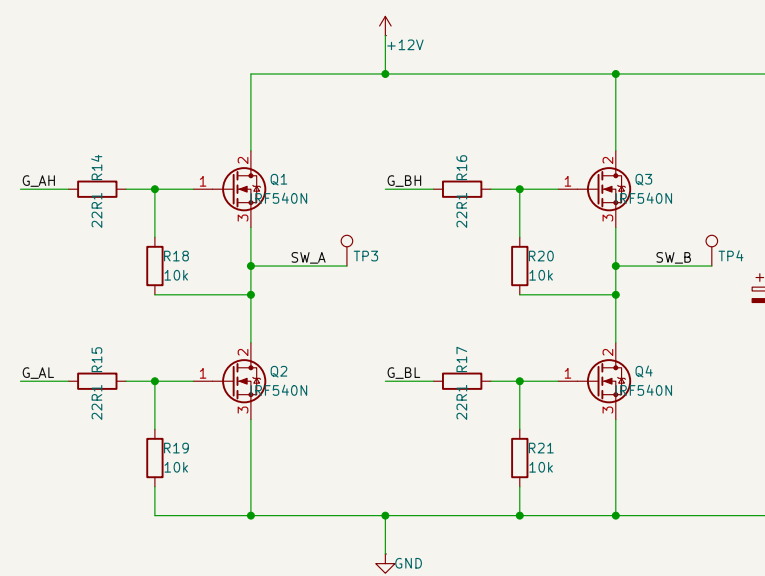
R12 = 3k3 on DEL gives -200 ns dead time (datasheet: 10k $\rightarrow$ 0.5 μ s, 100k $\rightarrow$ 4.5 μ s).

DIS is active high, so R13 holds it down.

No output DC blocking cap and no speaker ground – the load floats between the two legs.

L1/L2 are hand-wound on shop toroids: the stocked radial chokes saturate below the 3 A peak.

F Output bridge 4 x IRF540N



G Output filter + Zobel

